

1 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/bistable_dt10/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5p`, `ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

1.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/bistable_dt10`
- Number of saved time samples: 3241
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 1097 / 46
- RHS evaluations: 341537
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reltol} = 1.000 \times 10^{-6}$

1.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.007977, 165.628230] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-4.021 \times 10^6, 4.460 \times 10^7]$

1.3 Generated Artifacts

- Time-series CSV: `results/bistable_dt10/time_series.csv`
- Diagnostic summary JSON: `results/bistable_dt10/summary.json`
- Payload JLD2: `results/bistable_dt10/payload.jld2`

1.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

1.5 Included Plots

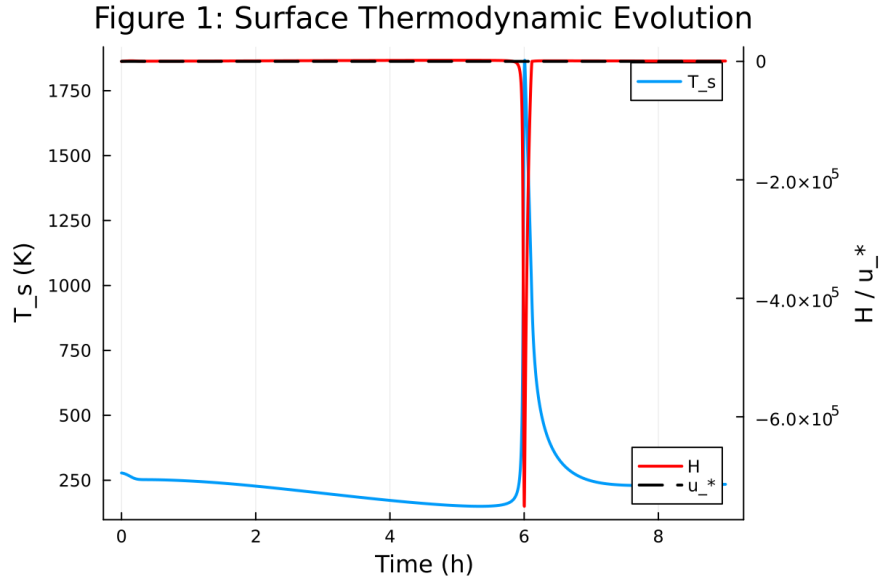


Figure 1: Time series of T_s , sensible heat flux, and friction velocity

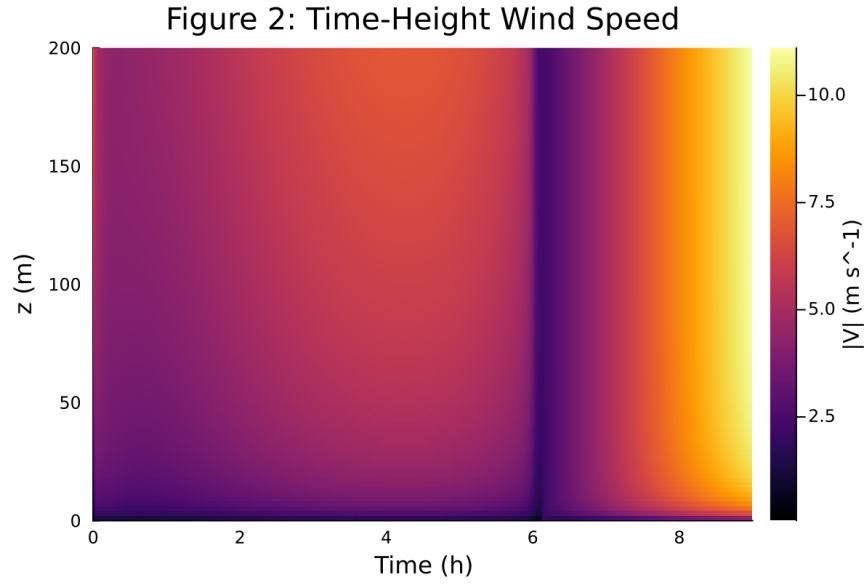


Figure 2: Time-height contour of wind speed with LLJ evolution

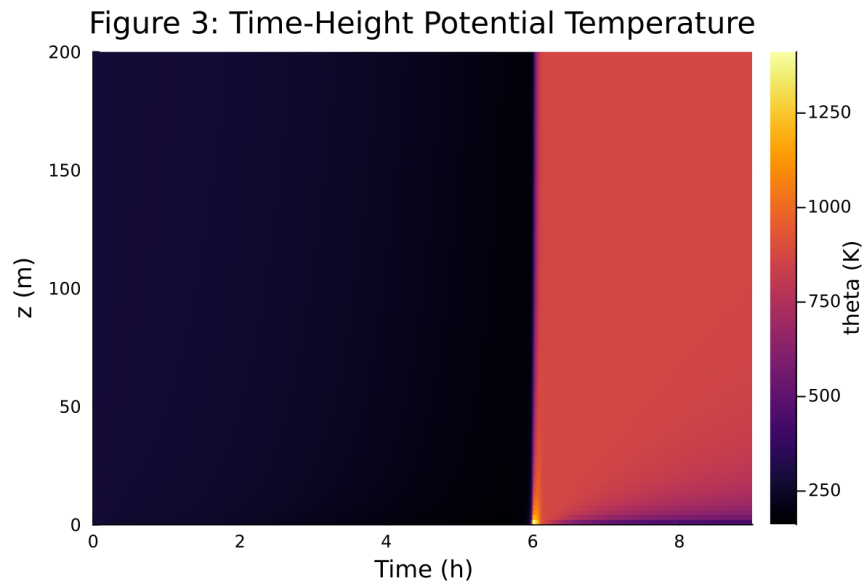


Figure 3: Time-height contour of potential temperature and inversion growth

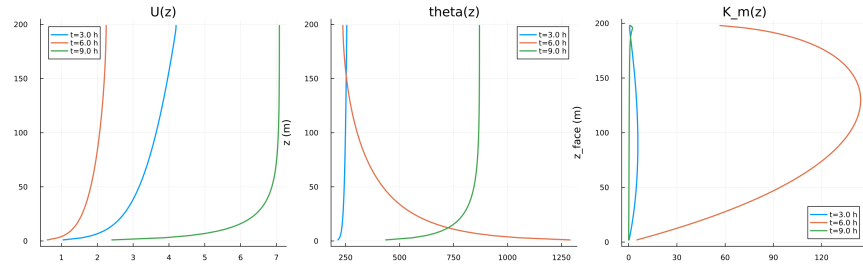


Figure 4: Vertical profiles of U , θ , and K_m at representative times

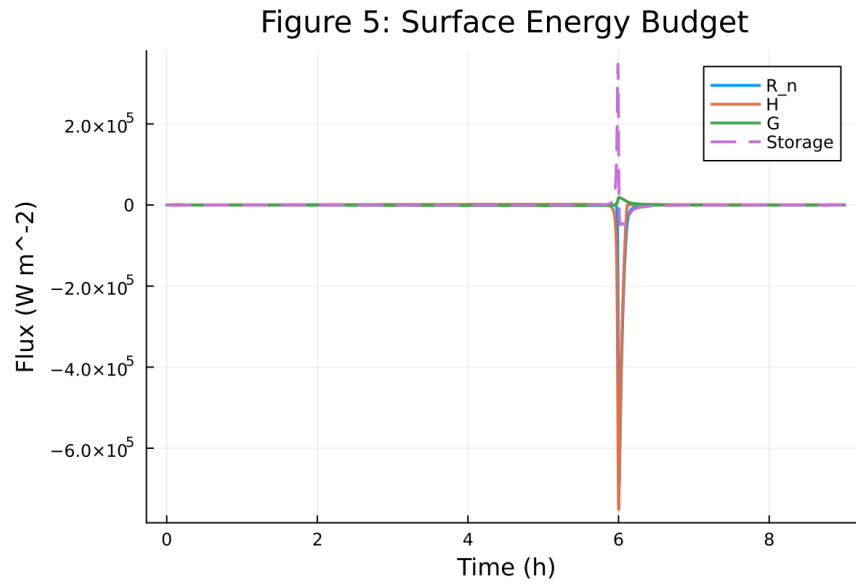


Figure 5: Surface energy budget components (R_n , H , G , storage)

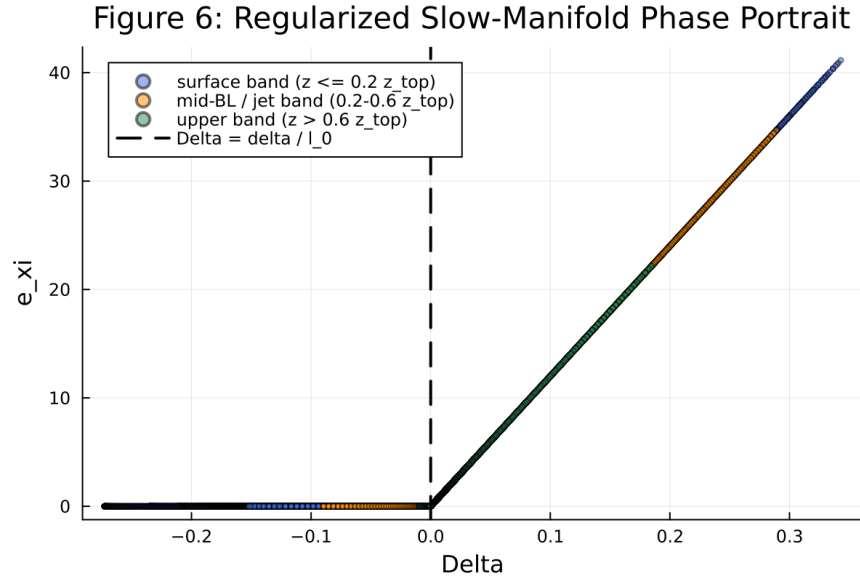


Figure 6: Phase portrait on regularized manifold (Δ vs e_{xi})

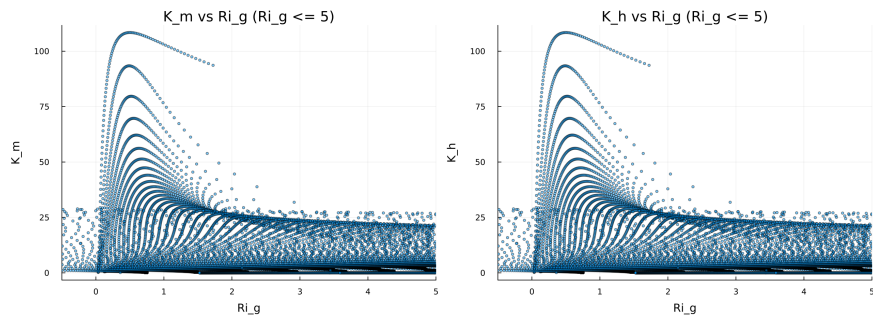


Figure 7: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

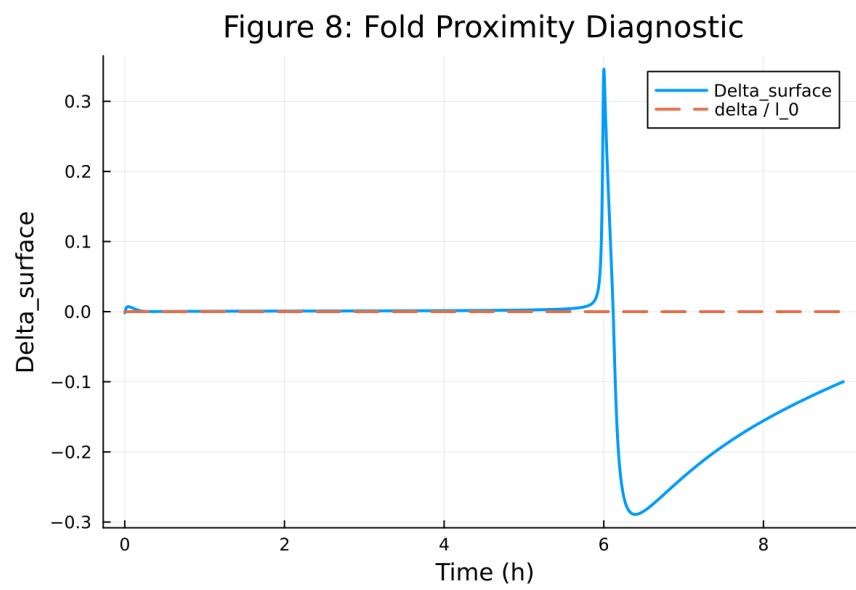


Figure 8: Fold proximity diagnostics versus time